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PROJECT: DESIGN THINKING ORIENTED PROPOSAL

THEME: AI TASK AUTOMATION SYSTEM

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INTRODUCTION

In this proposal, we use the **Design Thinking approach** to present an innovative **AI-powered solution** that can improve current systems in the industrial field. Our project focuses on three main phases — **Empathize**, **Define**, and **Ideate** — which will help us identify existing problems, understand user needs, and come up with effective solutions.

The chosen topic, **AI Task Automation System**, aims to solve one of the common challenges many industries face today — the heavy use of manual work in daily operations. Our idea is to create a system that uses artificial intelligence to automate repetitive and time-consuming tasks. By doing this, we hope to make industrial work more efficient, accurate, and less dependent on manual labour, while also encouraging companies to adopt more advanced technologies in their operations.

EMPHATIZE PROCESS

The Empathize phase is the first and most important stage of the Design Thinking process. It focuses on understanding the users and the challenges they might face before creating an innovative solution. For this project, our team began by conducting interviews with various industrial companies during the **Malaysia-Japan Industrial & Innovation EXPO (MJIIX2025)** to gain insights into the **current adoption of AI technologies** in real-world industries.

Through these interviews, we discovered that while many organizations recognize the potential of AI, they are still highly dependent on **manual, repetitive operations** in their daily workflows. These findings helped us better understand the underlying pain points in industrial environments and inspired our project idea - an **AI Task Automation System** that can automate repetitive manual tasks in both factories and offices.

Who are our users?

Our main users and stakeholders include:

- **Factory operators and machine technicians**, who perform repetitive operations, monitor production lines, and manage equipment schedules.
- **Engineers and supervisors**, who handle documentation, production reports, and quality monitoring.
- **Maintenance teams**, responsible for ensuring that machines run smoothly and troubleshooting issues when they occur.
- **Office administrators and HR personnel**, who manage document approvals, attendance, and inventory data manually.

- **Business owners and managers**, who oversee performance, cost efficiency, and overall productivity across departments.

These users represent different levels within an organization, yet they share a common goal — improving operational efficiency through smarter, automated systems.

What matters to them?

All these users value **efficiency, accuracy, productivity, and time management**. They want systems that help them complete their tasks faster, with fewer errors and less stress.

- **Factory operators** want to minimize manual input and reduce downtime caused by scheduling errors.
- **Engineers and supervisors** care about seamless data flow and faster approvals to keep projects on schedule.
- **Maintenance teams** value predictive systems that can alert them before a machine failure occurs.
- **Office administrators** prefer automated document handling to avoid confusion, delays, and redundant tasks.
- **Business owners and managers** are concerned with overall cost savings, long-term growth, and ensuring their companies stay competitive through digital transformation.

Ultimately, what matters to all users is a **smooth, automated workflow** that reduces manual effort, prevents human mistakes, and enables more strategic focus on higher-level work.

Problem and Pain

From our research and interviews, we identified several critical problems faced across industrial and administrative sectors:

1. **Repetitive manual work** - Employees spend excessive time on routine tasks such as document approvals, inventory updates, and scheduling.
2. **Human errors** - Manual data entry and communication gaps lead to mistakes that disrupt workflow and reduce productivity.
3. **Lack of automation tools** - Many companies have limited or no systems that can automate processes effectively.
4. **Low system integration** - Departments often work in isolation, causing inefficiencies and delays in coordination.

5. **Employee resistance to new technology** - Some staff members prefer traditional methods and are reluctant to adapt to AI-based systems due to lack of training.
6. **High operational costs** - Manual labor and inefficiencies increase costs for business owners and reduce profit margins.
7. **Poor data management** - Without centralized digital systems, it becomes difficult to track records, performance, and maintenance logs accurately.

For instance, one company mentioned that their entire approval process is still paper-based, requiring physical signatures that slow down communication. Another company stated that although they aim for 50% automation, they are struggling with workforce readiness and implementation strategies.

The Goal of the System

The goal of the Empathize stage is to deeply understand the real struggles of people working within industrial and office environments and identify how artificial intelligence can help solve them.

By putting ourselves in the user's position, we learned that they need a system capable of:

- **Automating repetitive manual tasks**;
- **Reducing human errors** through intelligent data handling;
- **Improving coordination and efficiency** between teams;
- **Freeing employees to focus on innovation and decision-making** rather than routine paperwork.

This understanding became the foundation for our proposed solution - the **AI Task Automation System** - an intelligent platform that integrates automation, learning, and data driven decision making. It directly supports the **Smart Industry** initiative by promoting digital transformation, reducing waste of time and human effort, and improving overall industrial efficiency.

DEFINE PROCESS

The Define phase builds on insights gathered during the Empathize stage to clearly identify and articulate the needs of users and stakeholders in industrial and administrative environments. By analysing the problems and pain points revealed through interviews, we can establish a focused understanding of what users require from an AI-powered automation system. This phase ensures that the proposed solution is aligned with real-world requirements and addresses both operational challenges and organizational goals.

Stakeholders and Users

According on our interview and research, we summarised the main stakeholder/user roles that might implement or be interested to implement the AI Task Automation System include:

User / Stakeholder	Role	Specific Needs
Factory Operators & Machine Technicians	Manage daily operations, monitor production lines, and operate machinery	Need automation for repetitive tasks, accurate scheduling tools, and alerts for maintenance to reduce downtime.
Engineers & Supervisors	Oversee documentation, project reporting, and quality control	Require seamless data flow, faster approvals, and accurate reporting to maintain project timelines.
Maintenance Teams	Ensure equipment reliability and troubleshoot issues	Need predictive alerts and automated monitoring to predict machine failures before they occur.
Office Administrators & HR Personnel	Handle document approvals, attendance, and inventory	Require automated workflows to reduce manual data entry, avoid errors, and streamline repetitive administrative processes.
Business Owners & Managers	Oversee overall performance, efficiency, and costs	Need actionable insights, cost-saving solutions, and operational transparency to support strategic decisions and digital transformation initiatives.

Identified Needs

With real examples that we gathered throughout our interview, we are proposing a list of needs across all groups that we have analysed:

a. Automation of Repetitive Tasks

- **Problem:** Employees spend too much time on manual, repetitive tasks.
- **Example:** T.T.E Engineering's document submission process is fully manual, requiring physical authorization, causing delays.
- **Need:** An AI system that automatically handles approvals, updates schedules, and manages routine workflows.

b. Error Reduction and Data Accuracy

- **Problem:** Manual data entry leads to mistakes that disrupt operations.
- **Example:** Office staff manually update inventory logs, often resulting in discrepancies that affect production planning.
- **Need:** Intelligent data handling to reduce errors and ensure accurate records.

c. Improved Workflow Coordination

- **Problem:** Departments work in isolation, causing delays.
- **Example:** Maintenance teams at Yanmar sometimes do not receive timely alerts about machine issues because operations and scheduling systems are disconnected.
- **Need:** A centralized AI system to integrate workflows, track tasks in real time, and improve communication.

d. Predictive and Proactive Support

- **Problem:** Unexpected machine failures or bottlenecks disrupt production.
- **Example:** NOK is beginning to explore predictive AI for future automation but currently lacks tools to anticipate issues.
- **Need:** AI-driven predictive alerts to prevent downtime and optimize resource allocation.

e. Ease of Adoption and Workforce Engagement

- **Problem:** Employees resist new technology due to lack of knowledge.
- **Example:** LabBase staff are comfortable with their current database but require guidance to leverage AI effectively.
- **Need:** User-friendly AI interfaces and gradual implementation strategies to encourage adoption.

f. Cost Efficiency and Strategic Value

- **Problem:** Manual operations increase operational costs.
- **Example:** Yanmar aims for 50% automation, but inefficient processes and errors raise costs and slow decision-making.
- **Need:** AI solutions that reduce labour intensive work, minimize errors, and provide measurable ROI for managers and owners.

IDEATE – AI SOLUTION

After analysing the key challenges and needs faced by industrial workers and administrators, the Ideate phase focused on generating innovative and practical ideas to address these issues through Artificial Intelligence. This stage made us go through brainstorming session between members to explore how AI could transform industrial workflows, reduce manual effort, and improve overall efficiency.

After talking with users and analysing their main challenges, our team came up with the idea of an **AI Task Automation System**. This system is designed to automate repetitive tasks, support smarter decision-making, and make industrial operations more efficient. During the ideation process, we focused on combining automation with AI intelligence while keeping the system practical and easy for real-world users to adopt.

Key AI-Driven Ideas Generated:

1. **AI-Powered Task Automation System:**
   A centralized platform that automates routine tasks such as report generation, scheduling, and document approvals. This reduces human workload, minimizes errors, and ensures faster task completion.
2. **Smart Workflow Assistant (NLP Chatbot):**
   An AI-powered assistant that interacts with employees using natural language processing (NLP). It guides users, answers operational queries, and provides real-time support to enhance workflow efficiency.
3. **Predictive Maintenance Module:**
   A machine learning model that continuously analyses equipment and sensor data to predict potential breakdowns. This enables proactive maintenance, reduces downtime, and minimizes repair costs.
4. **Automated Document and Data Management:**
   Utilizing Optical Character Recognition (OCR) and AI-based categorization to digitize manual approvals and data entry. It ensures quick retrieval, higher accuracy, and better data organization across departments.
5. **AI-Driven Performance Dashboard:**
   A real-time analytics dashboard that uses AI to track productivity metrics, detect inefficiencies, and provide actionable recommendations. This helps managers make data-driven decisions and improve overall operational performance.
6. **Intelligent Workforce Management:**
   AI algorithms optimize staff scheduling, shift allocation, and workload balancing across various departments. This ensures fair task distribution, prevents fatigue, and maximizes overall team productivity.

Through this ideation process, our team envisioned a smart industrial ecosystem that integrates automation, data intelligence, and human collaboration. Each AI-driven feature contributes toward solving the pain points identified during the Empathize stage, from reducing manual labour and human error to improving coordination and operational efficiency. The proposed solution supports the Smart Industry and Smart City initiative by enabling industries to transition into more efficient, data-driven, and sustainable workplaces.

GOALS OF AI SOLUTION

The **AI Task Automation System** is designed to address the real-world challenges faced by industrial and administrative users, as identified through interviews and research. Its goals are aligned with user needs, operational pain points, and the broader objective of promoting smart industry practices.

1. Automate Repetitive Tasks

Goal: Replace time-consuming manual operations with AI-driven automation. This includes tasks such as document approvals, report generation, scheduling, and inventory updates. By automating these repetitive processes, employees can focus on higher-value activities like problem-solving, innovation, and strategic planning, while reducing fatigue and increasing overall productivity.

2. Minimize Human Errors

Goal: Ensure precise and reliable handling of data and workflows through intelligent AI systems. This reduces mistakes caused by manual data entry, miscommunication, or inconsistent documentation, which can disrupt operations, delay production, and increase operational costs. Accurate data management enables better decision-making across all levels of the organization.

3. Improve Workflow Coordination

Goal: Integrate tasks and communication across departments to create a seamless, centralized workflow. This ensures that production, maintenance, engineering, and administrative teams work in sync, avoiding delays, misaligned schedules, and duplicated efforts. Real-time visibility and centralized tracking allow managers and teams to respond quickly to changing operational needs.

4. Enable Predictive and Proactive Decision-Making

Goal: Leverage AI to anticipate potential issues before they occur, such as machine breakdowns, maintenance needs, or bottlenecks in production. Predictive alerts and proactive recommendations minimize downtime, optimize resource allocation, and maintain continuous operations, ultimately improving efficiency and reducing costly interruptions.

5. Facilitate Workforce Adoption and Engagement

Goal: Ensure that AI systems are user-friendly, intuitive, and easy to integrate into daily workflows. Provide guidance, training, and gradual implementation strategies so employees of all skill levels feel confident adopting the technology. By addressing resistance to change, the system fosters a culture of innovation and maximizes the benefits of automation across the organization.

6. Enhance Cost Efficiency and Strategic Value

Goal: Reduce operational costs by minimizing manual labour, preventing errors, and improving resource utilization. Provide actionable insights through AI analytics and dashboards that help business owners and managers make informed, data-driven decisions. This supports long-term growth, competitiveness, and alignment with digital transformation initiatives, ensuring a measurable return on investment.

The overarching goal of the **AI Task Automation System** is to create a **smart industrial ecosystem** that combines automation, intelligence, and human collaboration. It aims to **reduce manual labour, minimize errors, improve coordination, and provide actionable insights**, enabling industries to operate more efficiently, sustainably, and strategically while supporting the broader Smart Industry initiative.

CONCLUSION

In this proposal, we have presented an AI Task Automation System designed to address the challenges faced by industries that rely heavily on manual, repetitive work. Through the Design Thinking process—emphasizing Empathize, Define, and Ideate—we gained a deep understanding of user needs, pain points, and operational inefficiencies in both industrial and administrative environments.

Our research highlighted key issues such as time-consuming manual tasks, human errors, poor workflow coordination, and resistance to adopting new technologies. By translating these insights into practical solutions, our AI-powered system aims to **automate routine operations, improve data accuracy, streamline workflows, enable predictive maintenance, and support workforce adoption**.

Ultimately, the proposed AI solution not only enhances productivity, efficiency, and cost-effectiveness for organizations but also encourages a shift toward digital transformation and smart industry practices. By implementing this system, companies can reduce manual labour, empower employees to focus on higher-value work, and position themselves for sustained competitiveness in an increasingly automated industrial landscape.